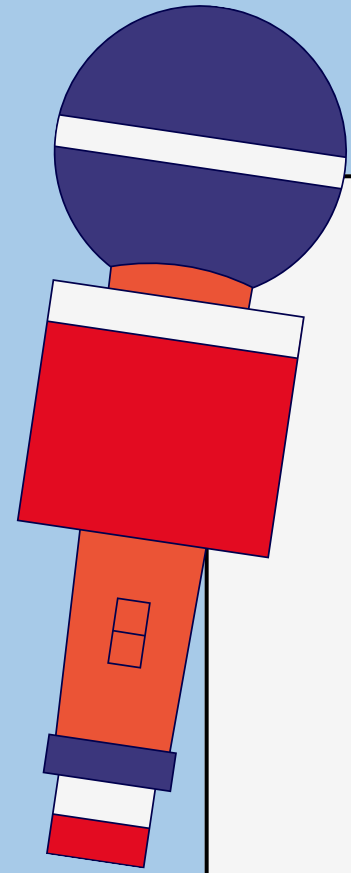




Web Crawlers using MalayMail.com

Group Shopping

NURUL ADRIANA BINTI KAMAL JEFRI (A23CS0258)

TAN YI YA (A23CS0187)

TEH RU QIAN (A23CS0191)



Objective

1. To design and deploy an asynchronous, effective web crawler to extract over 100,000 records of new articles across different categories.
2. To build multiple data processing pipelines using different frameworks to execute data cleaning, standardisation and meaningful structural transformation.
3. To compare and evaluate the performance of each optimisation framework using core metrics: execution time, CPU usage, memory consumption and throughput

Target Website

malaymail

Trusted digital site of Malaysia's oldest newspaper established in 1896, covering local and international news that matter to Malaysians.

Data to be Extracted

1. Article ID
2. Title
3. Category
4. Day
5. Date
6. Time
7. Author
8. Content
9. URL Link

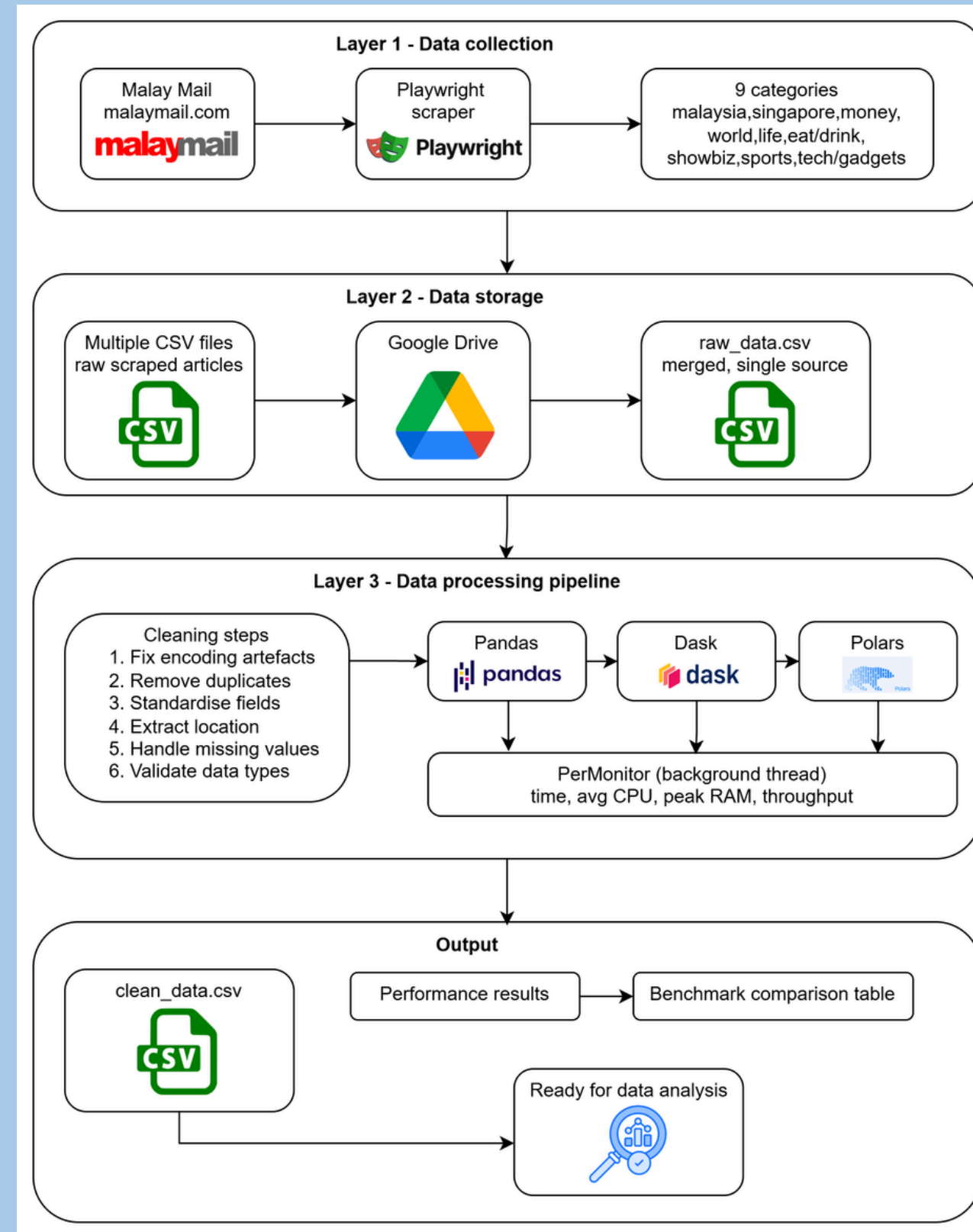
Aidiladha, Wesak, Kaamatan, Gawai: Malaysia's massive festive mash-up is about to take over



KUALA LUMPUR, May 24 — Malaysia is set to witness an “extraordinary festive atmosphere” soon as people of various races and religions celebrate several major festivals falling almost simultaneously during the school holiday season. The uniqueness of Malaysia’s diversity will be clearly on display as Muslims celebrate Hari Raya Aidiladha on...

FULL STORY ▶

System Architecture and Tools & Technologies Used

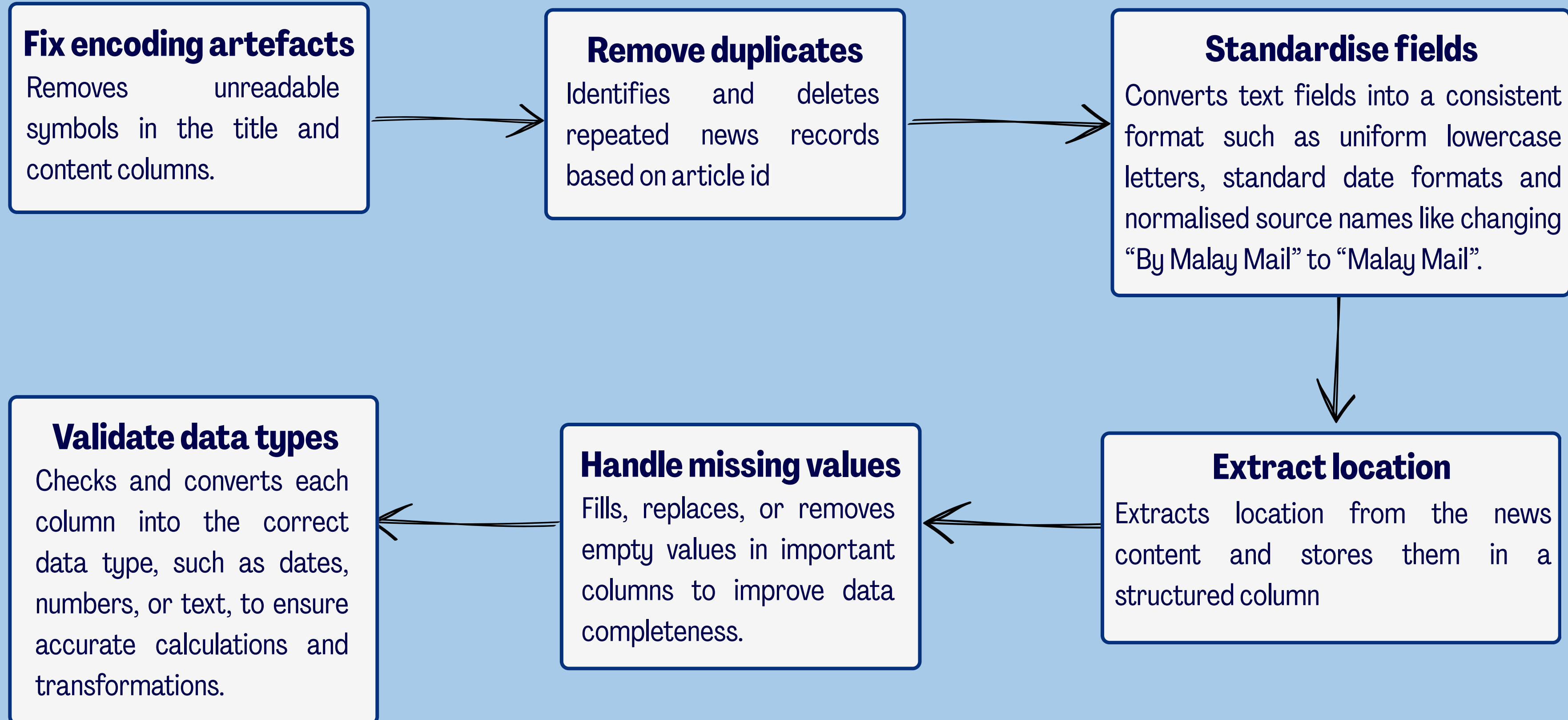


Raw data

- 121,174 news articles collected
- 9 columns
- ‘titles’ and ‘content’ columns with weird symbols and inconsistent formatting
- contains issues such as duplicate records, encoding problems, and missing values

id	title	category	day	date	time	author	content	link
220214	BAM president: Players flouting rules risk suspension	SPORTS	Saturday	16-May-26	8:20 PM MYT	By Malay Mail	KUALA LUMPUR, May 16 – National badminton players who violate Badminton Association of Malaysia (BAM) regulations face suspension from tournaments. BAM President Tengku Datuk Seri Zafrul Tengku Abdul Aziz said the proposed disciplinary measures aim to ensure players under the national governing body maintain high standards of discipline. –For those who do not meet the higher disciplinary standards we have proposed, appropriate action may be taken against them. –This includes barring them from competing (in certain tournaments), he told reporters after launching the Ampang Jaya Badminton Club (AJBC) hall today. It was reported recently that BAM Performance Committee Chairman Datuk Seri Lee Chong Wei had expressed grave concern over disciplinary issues plaguing national badminton players. Tengku Zafrul stressed that disciplinary problems were not new but must be addressed immediately if the national team is to regain its competitive edge at the highest level. He said BAM has already enforced several strict regulations, including making it mandatory for unmarried players to reside at the Academy Badminton Malaysia (ABM) hostel in Bukit Kiara. –One of the decisions made is to ensure all our players, including the seniors, stay together with the juniors at the ABM hostel, except for those who are married or have valid reasons, he added. – Bernama	https://www.malaymail.com/news/sports/2026/05/16/bam-president-players-flouting-rules-risk-suspension/220214

Data Cleaning



Cleaned data

- 118,284 rows
- 10 columns
- the location column has been added, weird symbols in the title and content columns have been removed,
- unnecessary prefixes like “By” in the author field have been cleaned.

id	title	category	day	date	time	author	location	content	link
220214	BAM president: Players flouting rules risk suspension	sports	Saturday	5/16/2026	8:20 PM MYT	Malay Mail	kuala lumpur	National badminton players who violate Badminton Association of Malaysia (BAM) regulations face suspension from tournaments. BAM President Tengku Datuk Seri Zafrul Tengku Abdul Aziz said the proposed disciplinary measures aim to ensure players under the national governing body maintain high standards of discipline. For those who do not meet the higher disciplinary standards we have proposed, appropriate action may be taken against them. This includes barring them from competing (in certain tournaments), he told reporters after launching the Ampang Jaya Badminton Club (AJBC) hall today. It was reported recently that BAM Performance Committee Chairman Datuk Seri Lee Chong Wei had expressed grave concern over disciplinary issues plaguing national badminton players. Tengku Zafrul stressed that disciplinary problems were not new but must be addressed immediately if the national team is to regain its competitive edge at the highest level. He said BAM has already enforced several strict regulations, including making it mandatory for unmarried players to reside at the Academy Badminton Malaysia (ABM) hostel in Bukit Kiara. One of the decisions made is to ensure all our players, including the seniors, stay together with the juniors at the ABM hostel, except for those who are married or have valid reasons, he added. Bernama	https://www.malaymail.com/news/sports/2026/05/16/bam-president-players-flouting-rules-risk-suspension/220214

Optimisation Techniques

Distributed Processing with Dask



Parallel Computing: Computations run in parallel using multi-processing, which reduces execution time for large datasets.

Partitioning: Datasets larger than memory are processed by small chunks and loaded as needed, which increases efficiency on disk-based datasets.

Dynamic Task Scheduling: Task graph known as Directed Acyclic Graph (DAG) is created to optimize and execute computations dynamically.

High Performance Data Frames: Polars



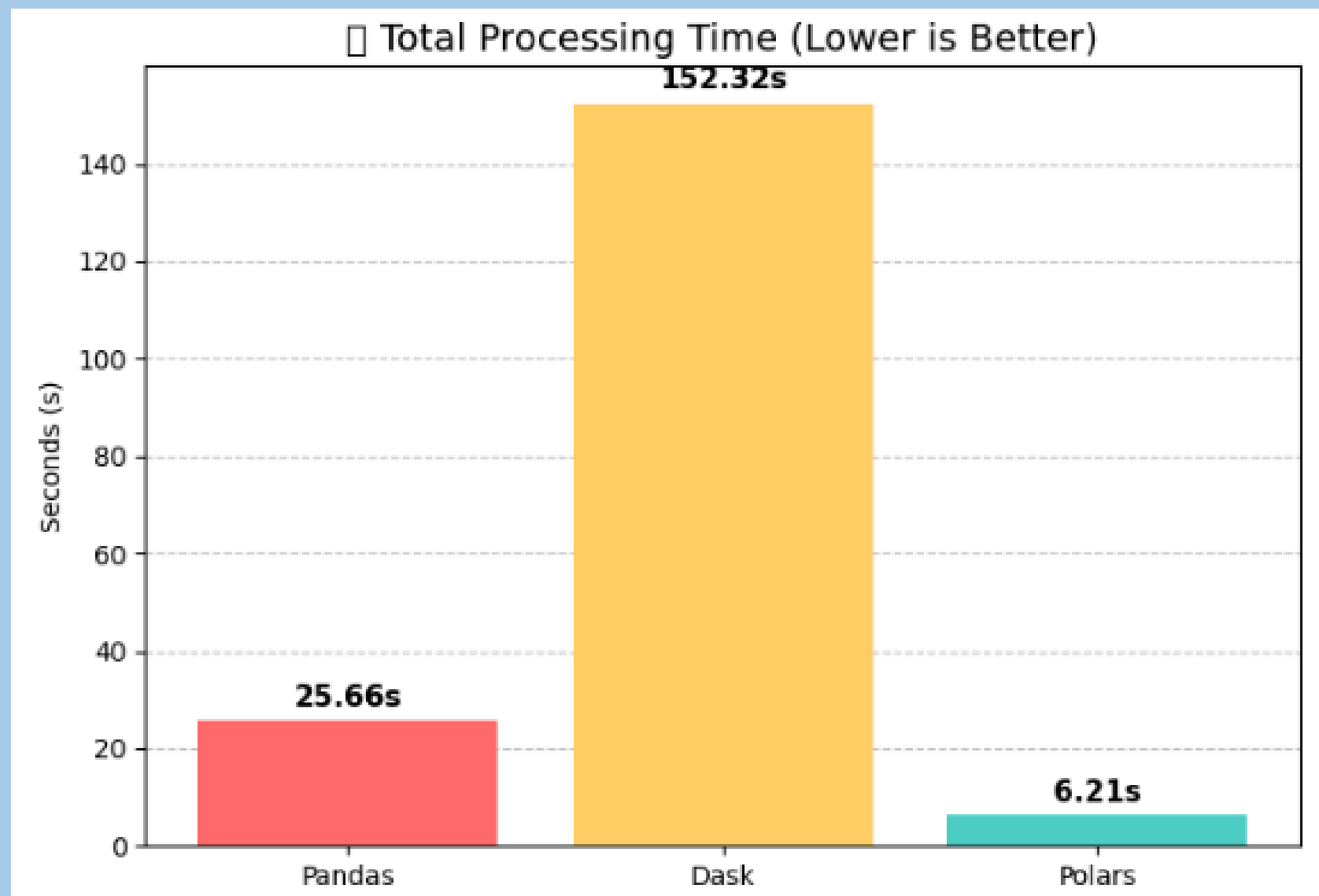
Built-in Parallelization: Does parallelize operations automatically. Data transformations are spread uniformly thus drastically decreasing execution time.

The Apache Arrow Columnar Memory Format: Offers cache coherent memory layouts and zero-copy data manipulation, completely avoids the huge overhead incurred.

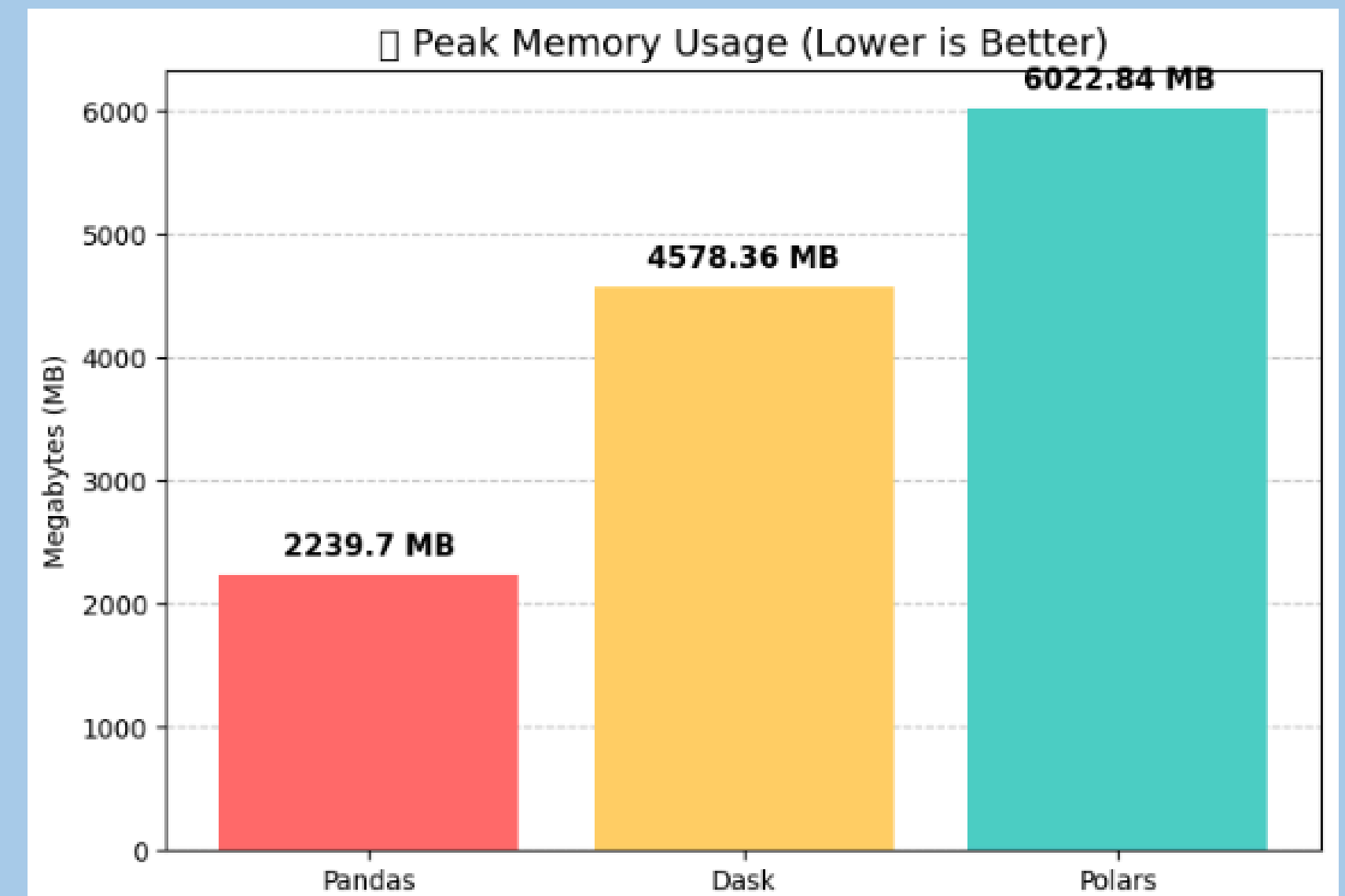
Expression Evaluation Engine: evaluates expressions sequentially in Rust, instead of executing one at a time while copying intermediate results to memory which helps cut through on data-cleaning jobs that involve more data.

Libraries Comparison

01 Total Processing Time

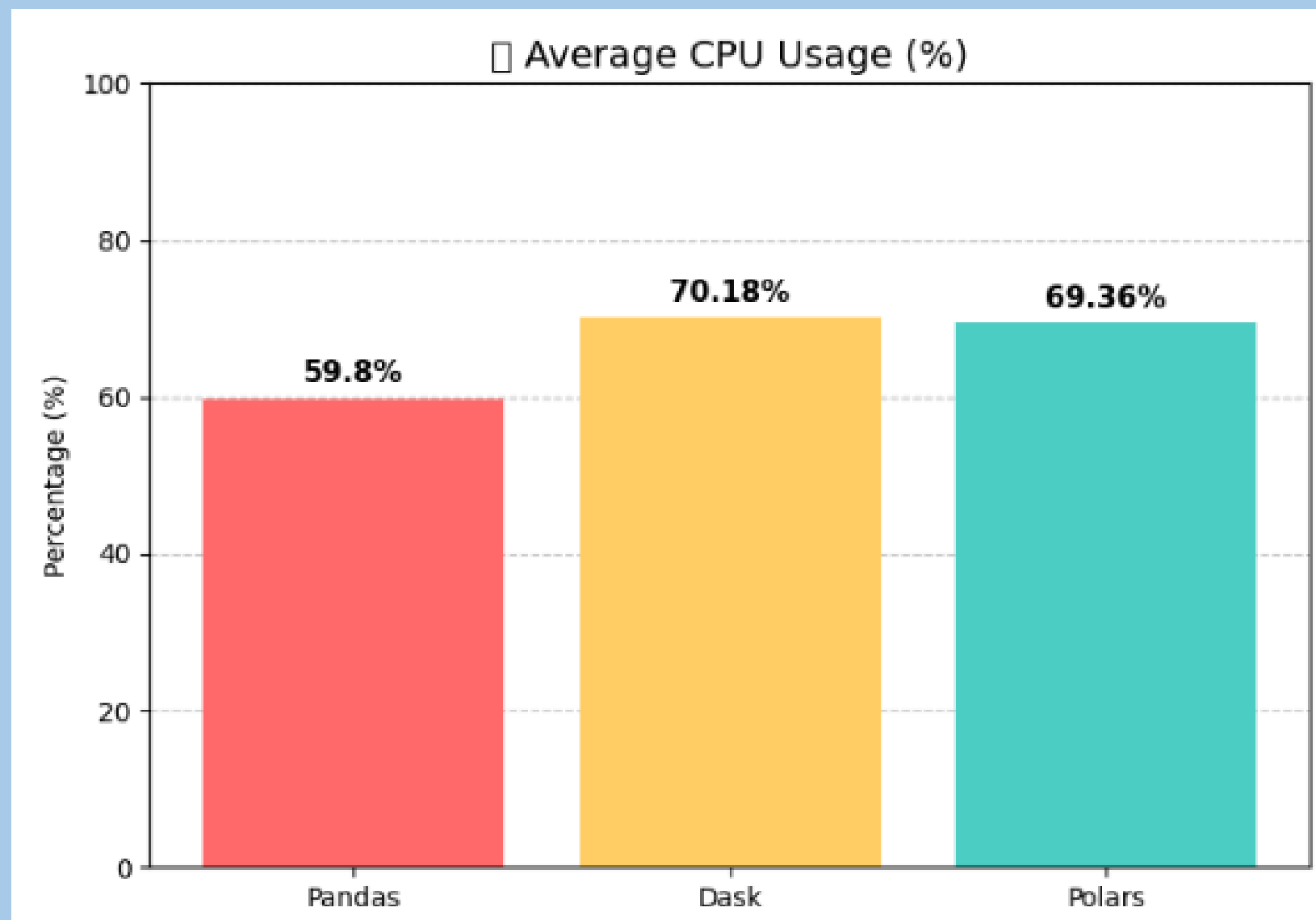


02 Peak Memory Usage

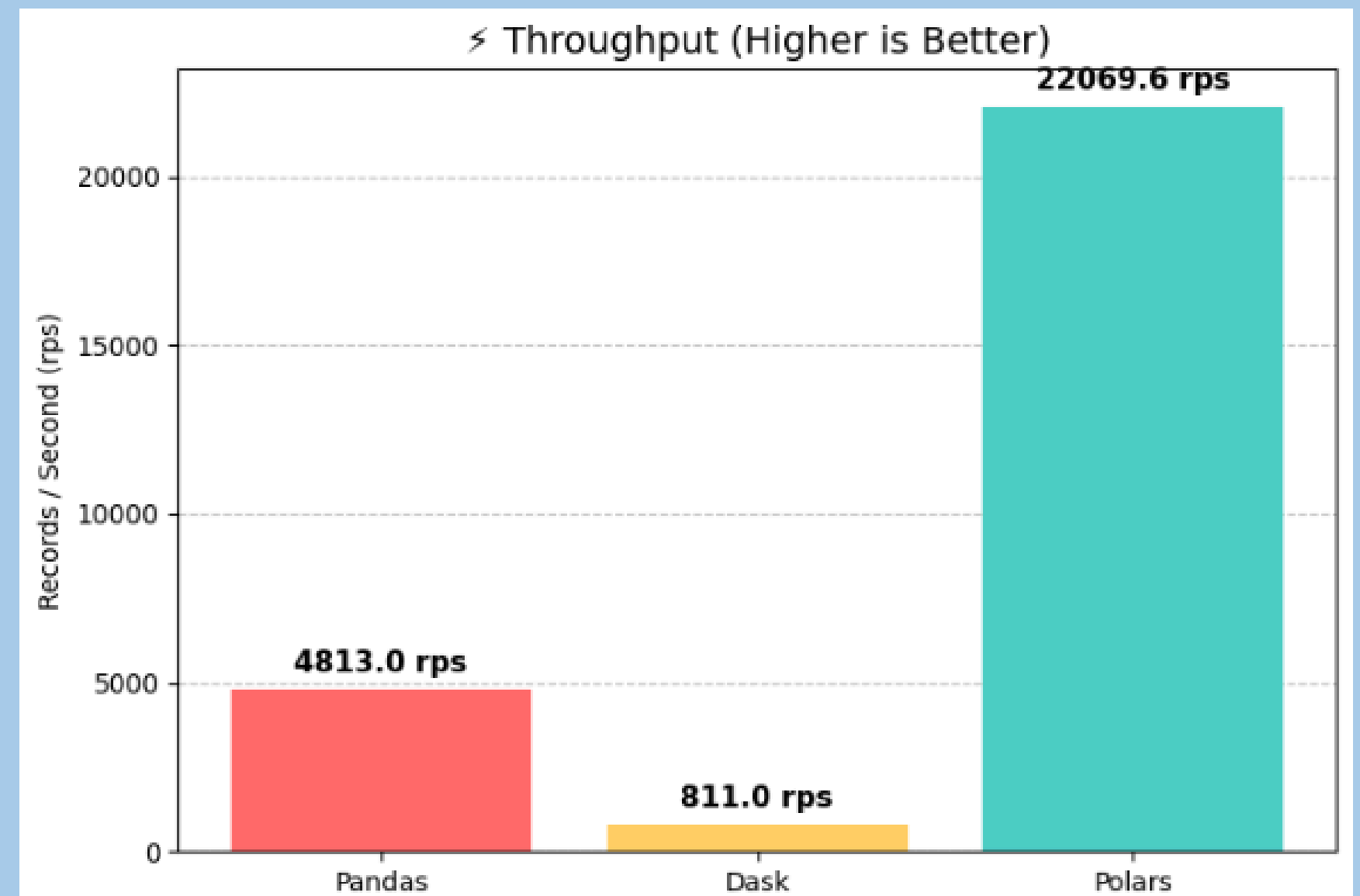


Libraries Comparison

03 Average CPU Usage



04 Throughput



Conclusion and Future Works

Polars is the most suitable framework for localized high-performance data engineering workloads due to superior speed and throughput



Future Improvements

- moving the current data processing setup to cloud platforms
- changing the pipeline from batch processing to real-time streaming system
- changing the storing format from csv to Apache Parquet